MILITARY SPECIFICATION

INDICATOR, ELECTRICAL TACHOMETER, ERU-12/A
0-350 RPM (ROTOR), 0-7000 RPM (ENGINE)

1. SCOPE

1.1 This specification defines a hermetically sealed, dual pointer, 0-7000 rpm and 0-350 rpm, helicopter tachometer indicator which is 3 inches in diameter and clamp mounted. (See 6.1 for suitable applications.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

DD-G-451	Glass, Flat and Corrugated, for Glazing, Mirrors, and Other Uses
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-665	Boxes, Paperboard, Metal Stayed Including Stay Material
PPP-B-676	Boxes, Set-up, Paperboard

Military

MIL-P-116	Preservation, Methods of
MIL-S-7742	Screw Threads, Standard, Optimum, Selected Series: General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-G-9398	Generator, Tachometer, High Temperature, Two-Pole, Aircraft, Type MB-6
MIL-L-25142	Luminescent Material, Fluorescent
MIL-M-26512	Maintainability Program Requirements for Aerospace Systems and Equipment
MIL-G-26611	Generator, Tachometer, GEU-7/A, Minature
MIL-R-26667	Reliability and Longevity Requirements, Electronic Equipment, General Specification for
MIL-D-70327	Drawings, Engineering and Associated Equipment

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-838	Lubrication of Military Equipment
MS3106	Connector, Plug, Electric, Straight
MS3108	Connector, Plug, Electric, Angle 90°
MS33558	Numerals and Letters, Aircraft, Instrument Dial, Standard Form of

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MS33585	Pointers, Dial, Standard, Design of Aircraft Instrument
MS33586	Metals, Definition of Dissimilar
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric Integral Mounting

(Copies of specifications, standards, publications, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification provides for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143.

3.2.1 Requirements conflict. In the event of a conflict, the requirements of this specification shall govern.

3.3 Materials

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life, unless protected by hermetic seal.

3.3.3 Dissimilar metals. Dissimilar metals are defined by MS33586 and shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion by means of protective coating or hermetic sealing.

3.3.4 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent or be protected by hermetic sealing.

3.3.5 Corrosive fumes. The materials as installed in the indicator and under the service conditions specified in 3.5 shall not liberate deleterious fumes.

3.3.6 Protective treatment. When materials are used in the construction of the indicator that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.4 Design and construction

3.4.1 The indicator shall be designed for operation from the varying-frequency alternating-current output of tachometer generators conforming to MIL-G-9398 or MIL-G-26611. The frequency of the generator output shall be proportional to the speed at which the generator is rotating. The design shall be such that with increasing and decreasing speeds the pointers shall move smoothly at a uniform rate consistent with the speed change. It shall be possible

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to operate both movements from one generator without affecting the operation of either movement.

3.4.1.1 The indicator shall be so constructed that no parts will work loose in service. The indicator shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The indicator shall be so constructed that adjustments and repairs can be easily made by the personnel of overhaul bases with tools and equipment normally available commercially.

3.4.1.2 Adjustment. Provisions for adjustment of the indicator shall not be based solely on changing the strength of the magnets.

3.4.2 Reliability. The indicator shall have a mean-time-between-failure (MTBF) of 750 hours. The reliability level shall be demonstrated to a 90 percent confidence factor.

3.4.3 Maintainability. Maintainability shall be in accordance with MIL-M-26512 and the following.

a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspections, et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive identification of the replaceable defective part, assembly, or component

c. Optimum accessibility in all systems, equipments, and components requiring maintenance, inspections, removal, or replacement

d. Sufficiently low mean time to accomplish corrective maintenance to assure the required operational availability of the equipment

e. Minimum number and type of spare parts and assemblies.

3.5 Performance. The indicators shall conform to the following requirements when subjected to the applicable tests of section 4.

a. Scale error. Scale error operation at room temperature, -54°C, -35°C, 0°C, and +71°C temperatures in accordance with tables I, II, III, and IV.

b. Friction error. Friction error not to exceed 40 rpm on the engine scale and 2 rpm on the rotor scale at tachometer generator drive speeds between 1000 and 4600 rpm

c. Position error. Position error not to exceed 25 rpm on the engine scale and 2 rpm on the rotor scale when indicator is positioned 45 degrees each way from the normal operating position with a tachometer generator drive speed of 4000 rpm. The normal operating position is with the longitudinal axis of the indicator horizontal and the dial in an upright position (word "engine" at the bottom).

d. Fogging and thermal shock. Subject the indicator to 85° ± 5°C following which the indicator shall be immersed in an ice water bath.

e. Vibration error. The error due to vibration shall not exceed 60 rpm on the engine scale and 3 rpm on the rotor scale, when the indicator is operating while vibrating in a plane inclined 45° to the horizontal in a circle having a 0.009 to 0.011 inch diameter at a

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TABLE I - ENGINE SCALE

GENERATOR DRIVE SHAFT SPEED	CORRECT INDICATED SPEED	SCALE ERROR TOLERANCE ±	SCALE ERROR TOLERANCE ± AT SPECIFIED TEMPERATURE			
		ROOM TEMP	-54°C	-35°C	0°C	+71°C
RPM						
0	0	50	100	90	80	70
500	770	70	100	90	80	70
1000	1540	70	100	90	80	70
2000	3080	70	120	90	80	80
2500	3850	70	120	90	80	80
3000	4620	70	120	90	80	80
3300	5082	50	120	90	80	70
3600	5544	50	120	90	80	70
3900	6006	20	120	90	80	60
4000	6160	20	120	90	80	60
4100	6314	10	120	80	70	50
4200	6468	20	150	90	80	60
4400	6777	40	150	100	80	70
4600	7085	60	150	100	80	90

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TABLE II - ENGINE SCALE

GENERATOR DRIVE SHAFT SPEED	CORRECT INDICATED SPEED	SCALE ERROR TOLERANCE ±	SCALE ERROR TOLERANCE ± AT SPECIFIED TEMPERATURE			
		ROOM TEMP	- 54°C	- 35°C	0°C	+ 71°C
RPM						
0	0	50	100	90	80	70
519	800	70	100	90	80	70
9740	1500	70	100	90	80	70
2013	3100	70	120	90	80	80
2532	3900	70	120	90	80	80
2987	4600	70	120	90	80	80
3311	5100	50	120	90	80	70
3636	5600	50	120	90	80	70
3896	6000	20	120	90	80	60
4026	6200	20	120	90	80	60
4091	6300	10	120	80	70	50
4220	6500	20	150	90	80	60
4415	6800	40	150	100	80	70
4610	7100	60	150	100	80	90

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TABLE III - ROTOR SCALE

GENERATOR DRIVE SHAFT SPEED	CORRECT INDICATED SPEED	SCALE ERROR TOLERANCE ± ROOM TEMP	SCALE ERROR TOLERANCE ± AT SPECIFIED TEMPERATURE			
R P M			-54° C	-35°C	0° C	+71°C
0	0	3	7	5	5	5
500	37.6	4	7	5	5	5
1000	75.3	4	7	5	5	5
2000	150.6	4	7	5	5	5
2500	188.2	4	7	5	5	5
3000	225.9	4	7	5	5	5
3300	248.5	4	7	5	5	5
3600	271.1	3	7	5	5	5
3900	293.7	2	7	5	4	3
4000	301.2	2	7	5	4	3
4100	308.7	1	7	5	4	3
4200	316.2	2	7	5	4	3
4400	331.3	3	7	5	4	4
4600	346.3	3	7	5	4	4

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TABLE IV - ROTOR SCALE

GENERATOR DRIVE SHAFT SPEED	CORRECT INDICATED SPEED	SCALE ERROR TOLERANCE ± ROOM TEMP	SCALE ERROR TOLERANCE ± AT SPECIFIED TEMPERATURE			
RPM			- 54° C	- 35° C	0°C	+ 71°C
0	0	3	7	5	5	5
531	40	4	7	5	5	5
1063	80	4	7	5	5	5
1992	150	4	7	5	5	5
2523	190	4	7	5	5	5
3055	230	4	7	5	5	5
3320	250	4	7	5	5	5
3586	270	3	7	5	5	5
3851	290	2	7	5	4	3
3984	300	2	7	5	4	3
4117	310	1	7	5	4	3
4250	320	2	7	5	4	3
4383	330	3	7	5	4	4
4649	350	3	7	5	4	4

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slowly varying frequency of from 5 to 50 cps.

f. Vibration failure. The indicator when mounted in the normal operating position shall withstand operation when vibrated for a 20-hour period in a plane inclined 45° to the horizontal in a circle having a 0.018 to 0.020 inch diameter at a frequency varied from 5 to 50 cps.

g. Temperature and altitude Subject to temperature ranging from -62° to +85° C without failure and altitudes to 52,000 feet. (Helium test)

h. Indicator starting. Starting and synchronous operation at an indicated speed of 200 rpm (engine scale) and 8 rpm (rotor scale) or greater with the following maximum voltages applied:

(1) Room temperature - 0.63 volts at 180 cycles per minute

(2) -54° C temperature - 0.72 volts at 180 cycles per minute

i. Magnetic effect. Subject the indicator, operating and nonoperating, rotated in increments of 45 degrees from normal, and positioned 5-1/2 inches from the center of a free-magnet compass having a field intensity of 0.17 to 0.19 oersted, without deflecting the compass more than 3 degrees.

j. Indicator current. The current drawn by each of the 3-phase windings shall not exceed 0.3 amp when connected to 21V 4,200 cpm supply.

k. Overspeed and reverse operation. Subjected to a generator shaft speed of 5,000 rpm for 5 minutes and then operated in the reverse direction at a generator-shaft speed of 2,500 rpm for 1 minute without exceeding a change in reading of 40 rpm on the engine scale and 2 rpm on the rotor scale.

l. Endurance. Subjected to 1000 hours of operation at an indication of 4000 ±500 rpm without failure or damage.

m. Salt spray. Subjected to salt spray for a period of 24 hours in accordance with MIL-STD-810 without evidence of external corrosion or deterioration affecting subsequent operation.

n. Insulation. Withstand the test voltages specified herein for a period of 5 seconds between connector pins A, B, and C and any metal part of the case or connector shell.

o. Sealing. Case leakage not to exceed 0.01 micron cubic foot per hour when the filling medium is nitrogen and helium, and 0.1 micron cubic foot per hour when the filling medium is 100 percent helium.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.7 Case. The case shall be in accordance with MS33639 for the 3-inch size. The length of the case shall not be more than 5.00 inches excluding the connector. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The case shall be so designed that the internal mechanism may be removed from the case, replaced, and the case resealed at least

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three times. This shall be accomplished without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.7.1 Material. The hermetically sealed case shall be constructed of a suitable material applicable to hermetic sealing, and the material shall be uniform in texture with a smooth external surface.

3.7.2 Finish. The portion of the indicator case exposed to view from the front of the mounting panel shall be finished in durable dull black. The finish applied to the remainder of the case shall be durable dull black.

3.7.3 Filling medium. The filling medium shall be at least 98 percent pure, free of dust particles, and contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or or a mixture of 88 percent to 92 percent nitrogen and the remainder helium as specified by the procuring activity (see 6.2). The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere.

3.8 Cover glass. The cover glass shall be in accordance with DD-G-451, type II quality AA. The minimum thickness of the glass shall be 0.093 inch. Any flaws allowed by DD-G-451 shall not interfere with the visibility of the dial.

3.8.1 Dial to cover glass distance. The distance between the inside surface of the cover glass and the indicator dial shall be the minimum practicable and shall not exceed 0.22 inch.

3.9 Dial. The dial shall consist of one flat surface. The dial material shall be suitable for the purpose and shall be subject to the approval of the procuring activity. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen nor turn when the indicator is vibrated. If screws are used for attaching the dial, they shall be so located that they will not interfere with any dial marking or prevent proper location of dial marking.

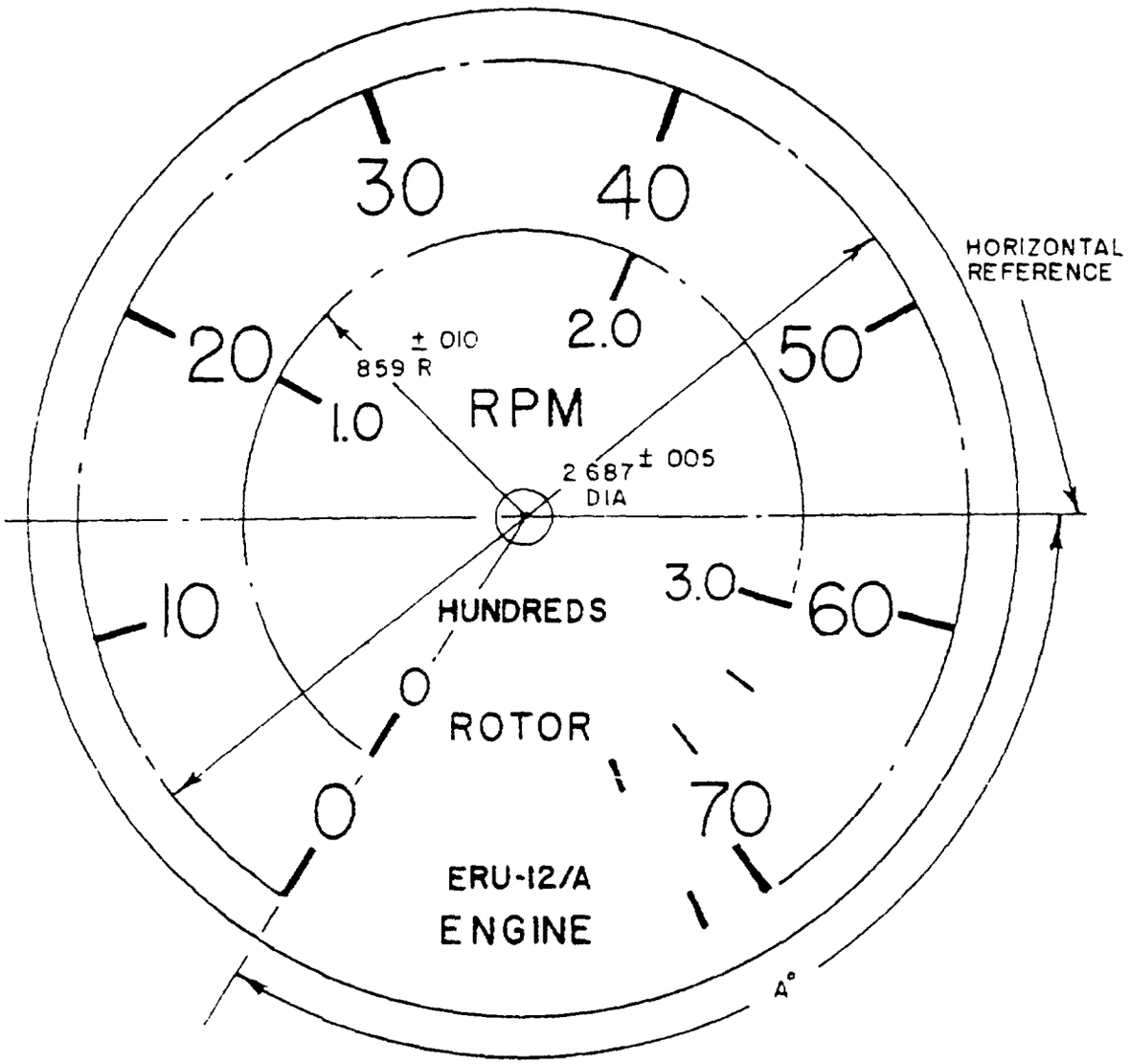
3.9.1 Markings. The dial markings shall conform to figure 1, figure 2, and table V and shall be durable enough to withstand environmental conditions specified herein. Unless otherwise specified, the style and proportions of numerals and letters on the dial shall conform to MS33558. The graduation to which each numeral applies shall be distinctly indicated. The dial graduations shall be uniformly spaced. The dial background and dimensions and colors of the markings shall be as specified in table V.

3.9.2 Dial visibility. Some portion of each marking on the dial of the indicator shall be visible from all points within a right angle truncated circular cone, the axis of which is the longitudinal axis of the indicator case, the periphery of the truncated end of which is the periphery of the aperture of the case of the indicator, and the side of which makes an angle of 30° with its axis.

3.9.3 Pointers. The pointers shall conform to figure 2. The length of the pointers shall be such that the tip of each pointer shall extend into its respective scale a distance equal to 1/5 to 2/3 of the length of the shortest graduation. The rotor pointer shall be the front pointer (nearest to the glass).

3.10 Electrical receptacle. Each indicator shall be provided with two hermetically sealed three-pin electrical receptacles conforming to MS33678-14S-7P. The receptacles shall

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OUTER SCALE	
NUMERAL MARKS	ANGLE A°
0	123
10	165
20	207
30	249
40	291
50	333
60	15
70	57
72	65.4

INNER SCALE	
NUMERAL MARKS	ANGLE A°
0	123
10	208 5
20	294
30	19 5
32	36 6
34	53 7
36	70 8

Figure 1. Dial Dimensions

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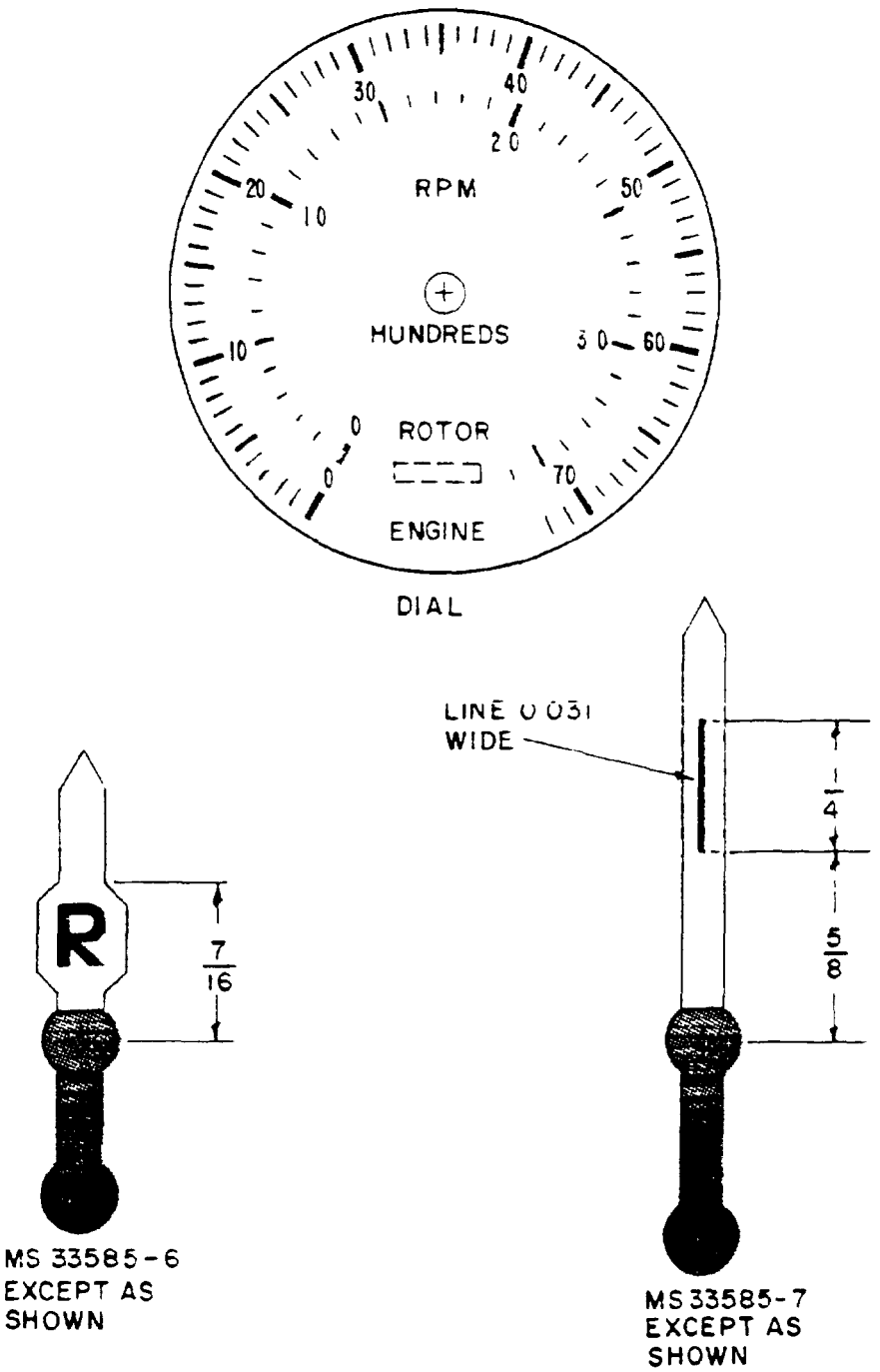


Figure 2. Dial And Pointers

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TABLE V - DIAL MARKINGS

MARKING	HEIGHT OR LENGTH ±0 016	WIDTH OF LINE OR GRADUATION ±0 005	COLOR
OUTER SCALE			
Graduations at numerals	0.156	0 030	
Major graduations between numerals	0.140	0 015	
Minor graduations	0.125	0 010	
Numerals 0, 10, 20, 30, 40, 50 60, and 70	0 172	0 029	
INNER SCALE			
Graduations at numerals	0.156	0 030	PER MIL-L-25142
Major graduations between numerals	0 094	0.015	
Minor graduations	0 070	0 010	
Numerals 0 1 0 2.0, 3 0	0 125	0 021	
Lettering "RPM"	0 156	0 026	
Lettering "HUNDREDS"	0 062	0.010	
Lettering "ROTOR" and "ENGINE"	0 094	0 015	
Letter "R" on pointer	0 1875	0 033	
Shaded portions of pointers	—	—	FED-STD 595
Dial background	—	—	
Line on engine pointer	0.25	0 031	No 37038
Lettering "ERU-12/A"	0 063	0.010	

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be hermetically sealed and located on a horizontal line through the center of the back of the case, with their centers 1-3/8 inches apart, with the polarizing key of the receptacle at the top. The receptacles will mate with electrical connectors conforming to MS3106-14S-7S or MS3108-14S-7S. Each receptacle shall be clearly labeled "engine" or "rotor", as appropriate. (See figure 3.)

3.11 Wiring. The three leads from each stator winding shall be designated as B, A, and C (conforming to phases 1, 2, and 3). The leads shall be connected to the B, A, and C pins of the electrical receptacles in such a manner that when 3-phase power with phase rotation sequence B-A-C is applied, the resultant indicator reading will be positive.

3.11.1 Electrical connections for counterclockwise generator drive shaft rotation (when viewed from the shaft end), for operating one or two indicators, shall conform to the wiring diagram shown on figure 3. For clockwise rotation, leads A and B shall be reversed at the generator plug.

3.12 Indication. The engine pointer shall indicate 6465 rpm and the rotor pointer shall indicate 316.2 rpm at a generator drive speed of 4200 rpm (see tables I, II, III, and IV for allowable tolerances). For a generator speed of 4600 rpm the pointer position shall be approximately 302° from the 0 rpm position.

3.13 Stop. The indicator mechanism shall be provided with a stop which, upon reverse operation of the indicator motor, shall stop the pointer motion after traveling in a CCW direction through a distance equal to but not less than 100-200 rpm on the engine scale.

3.14 Lubrication. Lubrication shall be accomplished in accordance with MIL-STD-838.

3.15 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.16 Weight. The weight of the indicator shall not exceed 2.35 pounds.

3.17 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.18 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Wiring, impregnation of coils, marking of parts and assemblies, welding, brazing, and painting shall be neat and thorough.

3.18.1 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they be shall held or limited accordingly.

3.18.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

3.18.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.18.4 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material after final assembly. Burrs and sharp edges as well as resin flash that may crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified

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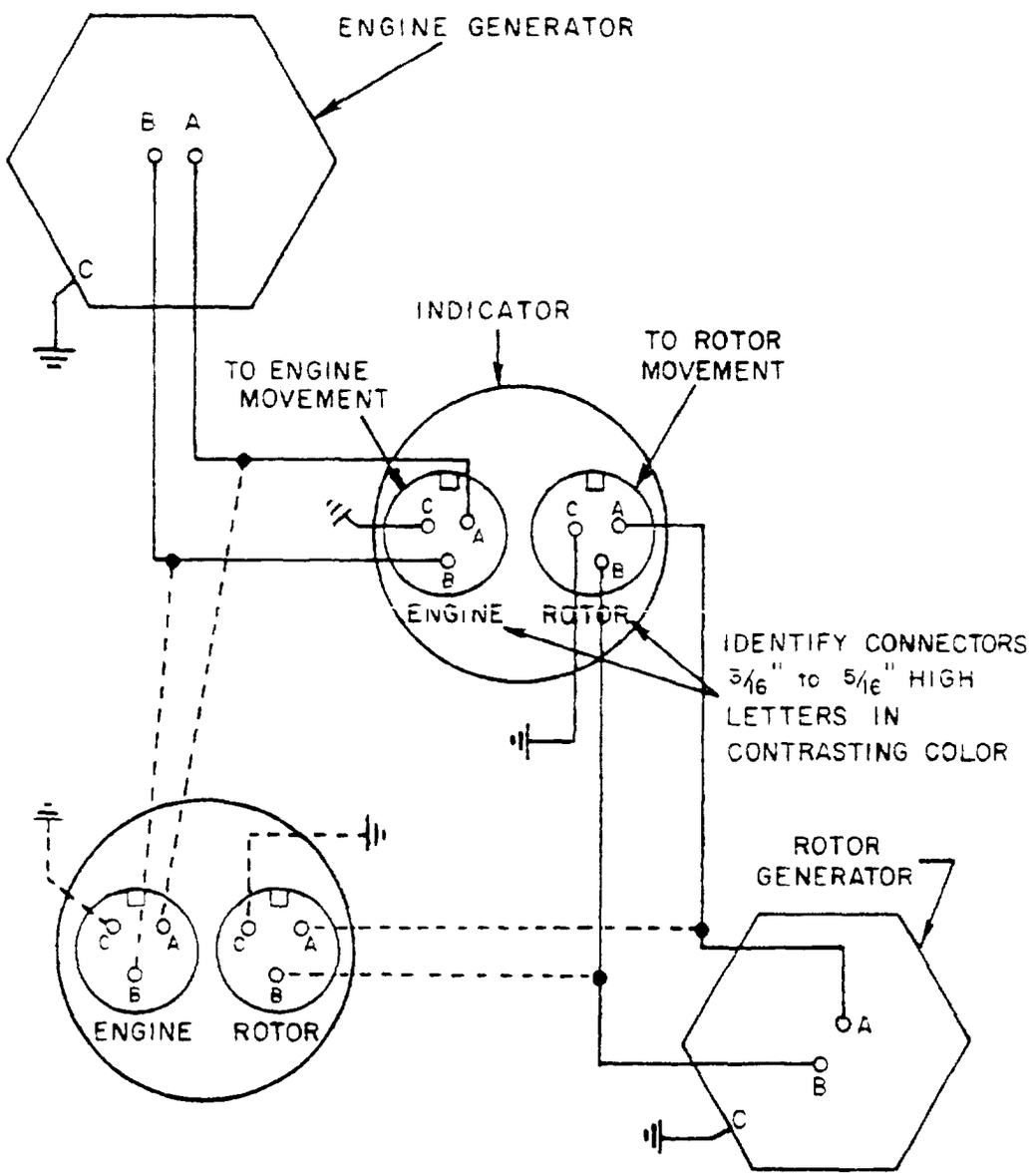


Figure 3. Wiring Diagram

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herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.3 Test conditions. The specific conditions for tests described under 4.6 are given in the following subparagraphs.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified conditions.

4.3.2 Tapping. Except where otherwise specified, the indicator may be gently vibrated or tapped before a test reading is taken.

4.3.3 Vibration. Unless otherwise specified, the indicator in its normal operating position shall be subjected to circular vibration in a plane inclined 45° to the horizontal plane. The vibration shall be within a frequency range of 5 to 50 cps and at amplitudes within 0.009 to 0.020 inch.

4.3.4 Attitude. Unless otherwise specified, the indicator shall be tested in its normal operating position.

4.3.5 Test voltage. Unless otherwise specified, voltages for indicator operation noted herein are alternating current root-mean-square values. The voltmeter used to measure alternating voltages shall be of a type which is free from frequency errors.

4.3.6 Tachometer generator. Wherever the words tachometer generator or generator are used in this specification it shall mean a tachometer generator conforming to MIL-G-26611 or to MIL-G-9398. If, as for the starting test, the voltage output of the generator is too high, resistors shall be used to reduce the voltage.

4.3.7 During the following tests, both tachometer mechanisms may be connected to the same generator if desired: Magnetic effect, overspeed and reverse operation, vibration error and vibration failure.

4.3.8 Testing. Unless otherwise specified, the following tests shall be conducted with the rotor and the engine mechanisms operating from separate generators. The mechanism not being tested shall be operated with a generator speed of 900 ±100 rpm while the other mechanism is being tested.

a. Friction error

b. Position error

c. Vibration error

d. Room temperature scale error.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of 10 indicators representative of the production equipment.

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Three of the indicators shall be subjected to all the tests except reliability, and the other 7 shall be subjected to the reliability test only. The samples shall be identified by the manufacturer's own part number, serial number, and such other information as required by the procuring activity. Test samples shall be referred to by serial number in all test data submitted to the procuring activity.

4.4.2 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of the following:

a. Individual tests

b. Sampling plans and tests.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Room temperature scale error

c. Friction error

d. Fogging

e. Room temperature starting

f. Sealing.

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. One indicator shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. High temperature exposure and scale error

c. Low temperature exposure, starting, warmup and scale error

d. Vibration error

e. Test for helium

f. Dielectric strength.

4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2) 3 indicators shall be selected at random from the first 15 on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Position error

c. Magnetic effect

d. Indicator current

e. Overspeed and reverse operation

f. Vibration failure

g. Endurance

h. Salt spray.

Note: See 4.3.7 for order of testing.

4.5.2.3 Sampling plan C. Seven indicators shall be selected at random by a government inspector from the first 15 produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Reliability.

4.5.2.4 Rejection and retest. When an item from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected.

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The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.4.1 Individual tests may continue. For production reasons, individual tests or sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. Each indicator shall be examined to determine compliance with the applicable drawings and to all the requirements of this specification, not covered by tests. Specific areas checked shall include the dimensions of the dial markings and case, workmanship, materials and dial visibility.

4.6.2 Scale error at room temperature. The indicators shall be connected as shown in figure 8 to a generator and tested at points of the scale indicated in tables I, II, III, and IV. The test shall be made with speeds increasing then with speeds decreasing. With speeds increasing, the speed shall be brought up to but shall not exceed the speed specified to give the desired reading, and with speeds decreasing, the speed shall be brought down to but shall not fall below the speed specified to give the desired reading. The errors at the test points shall not exceed the tolerances specified in the applicable table. When the speed is held constant at any point of the scale, the pointers shall not oscillate over a range greater than 40 (±20) rpm on the engine scale and 2 (±1) rpm on the rotor scale.

4.6.3 Friction error. The indicator shall be operated at generator speeds of 1000, 2000, 3000, 3700, and 4400 and the indication before and after vibration shall be noted. The speed shall be brought to, but shall not exceed the speed specified to give the desired reading. The indicator shall be clamped in the normal operating position, and no reading shall be taken until the pointer has stabilized. The change in indication before and after vibration shall not exceed 40 rpm on the engine scale and 2 rpm on the rotor scale.

4.6.4 Position error. The indicator shall be operated at 4000 rpm with the indicator in the normal (dial vertical) position. The indicator reading shall be noted. The maximum deviation from the normal position reading, when tested in the following positions, shall not exceed 25 rpm on the engine scale and 2 rpm on the rotor scale:

a. Rotated from normal position through 45 degrees to the rear (dial tilted 45 degrees back from vertical position)

b. Rotated from normal position through 45 degrees to the front (dial tilted 45 degrees forward from vertical position)

c. Rotated from normal position through 45 degrees to the right (dial vertical)

d. Rotated from normal position through 45 degrees to the left (dial vertical).

4.6.5 Fogging. The indicator shall be placed in a controlled ambient temperature of 85° +5°C for a minimum period of 2 hours. Within 20 seconds after removal from the oven, the indicator shall be immersed to a depth of

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1/2 ±1/4 inch in an ice water bath maintained at 0° to 7°C. After 3 minutes the indicator shall be removed from the fluid bath and quickly wiped dry (compressed air shall not be used). The cover glass and dial shall immediately be inspected for evidence of fog or damage. Evidence of fogging or damage shall be cause for rejection.

4.6.6 Thermal shock. The indicator shall be subjected to 3 cycles of exposure to water at 85° ±5°C and 0° ±7°C without evidence of moisture penetration or damage to coating or enclosures. Each cycle of the test shall consist of immersing the indicator in water at 85°C for a period of 30 minutes and then, within 5 seconds of removal from the bath, immersing the indicator for a period of 30 minutes in a bath maintained at 0° +7°C. This cycle shall be repeated continuously until 3 cycles have been completed. The sealing test, 4.6.18, shall be conducted after this test.

4.6.7 High temperature exposure and scale error at high temperature. The indicator shall be placed in a chamber and the internal temperature of the chamber raised to 71° ±2°C with the relative humidity not more than 5 percent. The source of heat for the chamber shall be arranged in such a manner that radiant heat will not fall upon the test specimen. The total volume occupied by a single item of equipment or by several items shall not exceed 50 percent of the internal volume of the test chamber. The indicator shall be maintained at 71° ±2°C for a period of at least 20 hours. While still at this temperature, the indicator shall be subjected to a scale error test (4.6.2) for increasing speeds only, without exhibiting sticking or erratic characteristics. After the chamber has returned to room temperature and the indicator has remained at room temperature for a minimum period of 4 hours, it shall be subjected to the scale error at room temperature test (4.6.2) increasing speeds only, and the friction test (4.6.3). No damage shall result that would adversely affect subsequent operation.

4.6.8 Low temperature exposure, starting, warmup and scale error at -54°C, -35°C, -0°C.

4.6.8.1 The indicator shall be subjected to a temperature of -62° ±3°C for a minimum period of 20 hours. No voltage shall be applied to the indicator during this time. The indicator shall then be maintained at a temperature of -54° ±3C for a minimum period of 3 hours. No voltage shall be applied to the indicator during this period or prior to the starting test. While at -54°C the indicator shall be subjected to the starting test, 4.6.9, except that the voltage applied shall be a maximum of 0.004 times the frequency in cycles per minute.

4.6.8.2 While still at the -54°C temperature, with no voltage having been applied for the preceding 15 minutes, both the engine and rotor mechanism shall be instantaneously connected to a tachometer generator operating at 4400 rpm. Within a period of 10 seconds the engine pointer shall read 6890 ±150 rpm and the rotor pointer shall read 331 ±6 rpm. The indicator shall not be operated for more than 10 seconds during this test.

4.6.8.3 While still at -54°C, with no voltage having been applied for the preceding 15 minutes, the scale error test, increasing and decreasing values, shall be conducted.

4.6.8.4 The indicator shall then be maintained at a temperature of -35° ±3°C for a minimum period of 3 hours. No voltage shall be applied to the indicator during this time. While still at -35°C the scale error test, using both increasing and decreasing values, shall be conducted.

4.6.8.5 The indicators shall then be maintained at a temperature of 0° ±3°C for a minimum period of 3 hours. No voltage shall be applied to the indicators during this time. While still at 0°C the scale error test, using both increasing and decreasing values, shall be conducted.

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4.6.9 Starting room temperature. One mechanism of the indicator shall be connected to a suitable tachometer generator through a variable resistance device. The voltage supplied to the indicator shall not be greater than 0.0035 times the frequency in cycles per minute. The frequency of the power supply shall be increased from 0, and the speed at which the indicator starts to run at synchronous speed shall be noted. This frequency shall not exceed 180 cycles per minute. The rotor mechanism and the engine mechanism shall be subjected to this test. At 180 cpm the engine pointer shall read at least 200 rpm and the rotor pointer shall read at least 8 rpm. The indicator may be gently vibrated during this test.

4.6.10 Magnetic effect. The indicator, first not operating and then operating at an indication of approximately 6000 rpm engine scale and 300 rpm rotor scale (both mechanisms may be connected to the same generator), shall be rotated in a vertical plane about a short bar magnetic compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 3°.

4.6.11 Indicator current. The current drawn by any one of the 3-phase windings shall not exceed 0.30 amp when the indicator is connected to a 21V 4,200 cpm power supply.

4.6.12 Overspeed and reverse operation. Both indicator mechanisms shall be electrically connected to a tachometer generator. The readings of the indicator shall be determined when the generator is operated at 1000 rpm and 4400 rpm. The indicators shall be at room temperature and the readings shall be taken before the indicators have attained an appreciable temperature rise. The generator and indicators shall then be operated, first at a generator shaft speed of approximately 5,000 rpm for a period of 5 minutes, and then in the reverse direction of rotation at a generator drive shaft speed of approximately 2,500 rpm for a period of 1 minute. After the indicators have been allowed to cool to room temperature, the readings of the indicator shall again be determined when operated at 1000 rpm and 4400 rpm. The change in indicated speed at the respective scale points before and after the overspeed test shall be not more than 40 rpm on the engine scale and 2 rpm on the rotor scale.

4.6.13 Vibration error. The indicator reading shall be determined when the generator is operated at a point between 3000 and 3700 rpm as selected by the inspector. The indicator shall then be subjected to circular vibration as specified under test conditions at an amplitude of 0.009 to 0.011 inch diameter. The indicator readings shall be determined with the generator operating at the original speed as the vibration frequency is varied from 5 to 50 cps. The indicator shall be in the normal operating position. The readings with vibration shall not vary more than 60 rpm on the engine scale and 3 rpm on the rotor scale from the readings without vibration. The amplitude of the pointer oscillation shall not exceed 60 rpm on the engine scale and 3 rpm on the rotor scale.

4.6.14 Vibration failure. The indicator shall be mounted on a vibration stand in its normal operating position. The test shall be conducted while the indicator is being operated at an indication of

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approximately 6000 rpm. The indicator shall be subjected to vibration with a circle diameter between 0.018 and 0.020 inch, at a frequency of 50 $^{+5}_{-0}$ cps for a 20-hour period. Once each hour during this test the frequency shall be varied uniformly from 50 to 5 cps and return. This shall be accomplished in 5-10 minutes. After the 50-hour period, the indicator shall be subjected to the increasing speeds portion of the scale error at room temperature test. The reading after vibration shall not differ from the reading before vibration by more than 35 rpm on the engine scale and 2 rpm on the rotor scale. In addition, no looseness in the mechanism or damage to any part of the instrument shall result from this test.

4.6.15 Endurance. The scale error test at room temperature shall be conducted. The indicator shall be operated at a generator speed of 4000 ±500 rpm for a period of 1000 hours. Both indicator mechanisms shall be operated simultaneously from two generators. The voltage at the indicator terminals shall be 18.5 ± 1V during the 1000-hour period. At the end of this period and after the indicator has cooled to room temperature, the scale error test at room temperature shall again be conducted. Failure to meet the 71°C temperature scale error tolerances as specified in table I, II, III, or IV shall be cause for rejection.

4.6.16 Salt spray. The indicator shall be subjected to a salt fog test in accordance with method 509.1, procedure I of MIL-STD-810, except that no pretest data will be required, and the test duration shall be 24 instead of 48 hours.

4.6.17 Dielectric strength. This test to be conducted after the salt fog test. A potential at a commercial frequency shall be applied between pin A, B, or C of each electrical receptacle and any part of the case or the electrical receptacle shell for a minimum period of 5 seconds. The applied voltages shall be 200V for indicators using the helium filling medium and 400V for the nitrogen-helium mixture filled indicators. There shall be no breakdown of insulation. The external surface of the indicator may be washed with distilled water and air dried prior to conduction of the dielectric strength test.

4.6.18 Sealing. The instrument case shall be tested for leaks by means of a mass spectrometer type of helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.01 micron cubic foot per hour at a pressure differential of at least 9/10 of an atmosphere. Where 100-percent helium is utilized as the filling medium, the leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of at least 9/10 of an atmosphere.

Note: The sealing test also tests the structural integrity of the indicator up to 51,000 feet altitude.

4.6.19 Helium test. This test shall be performed on one indicator selected at random from the three test samples not used for reliability testing. The indicator case shall be punctured, and the indicator subjected to the test for helium by means of a mass spectrometer type helium detector. Failure to detect helium shall be cause for rejection.

4.6.20 Reliability. During preproduction testing, indicator reliability shall be demonstrated in accordance with MIL-R-26667. Sever indicators shall be subjected to a reliability test to demonstrate a MTBF of 750 hours to a 90 percent confidence level. The level of reliability shall be verified when the indicators have operated for a minimum of 1,790 X (T) hours with a maximum of M failures. T and M are defined in table VI.

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TABLE VI Accept-Reject Criteria

M (Allowable Number of failures to the specified T)	T
0	4.4
1	5.2
2	6.0
3	6.8
4	7.6
5	8.5
6	9.3
7	10.1
8	10.9

4.6.20.1 Performance characteristics. The performance characteristics to be measured during the reliability test shall consist of a scale error test at room temperature, for increasing speeds only. Failure of either indicator mechanism of any reliability test sample to meet the scale error test shall be a failure of that indicator. The allowable error shall be as specified for +71°C in tables I, II, III, and IV for the engine scale and rotor scale.

4.6.20.2 Test level. The test level shall be in accordance with MIL-R-26667, test level VI except that the indicators will be operated continuously at a generator drive shaft speed of 3000 - 4000 rpm. Input voltages need not be monitored.

4.6.20.3 Preventive maintenance. No preventive maintenance may be performed.

4.6.20.3.1 Accept-reject criteria. The accept reject criteria is defined in table IV, of this specification.

4.6.20.3.2 Performance checking. The performance of the equipment (scale error test as specified in 4.6.2 (increasing speeds only) shall be conducted not less than once during each 66 hours of operation. The longevity test of MIL-R-26667 does not apply.

4.6.21 Maintainability verification. Maintainability verification shall be in accordance with MIL-M-26512, appendix A, paragraph entitled "Maintenance Task Test and Demonstration." All tasks selected shall be effectively simulated.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. Each indicator shall be preserved method 1A of MIL-P-116 and packaged in accordance with MIL-P-7936, one each in a container conforming to PPP-B-665, PPP-B-566, PPP-B-676 or PPP-B-636. Electrical connectors shall be provided with protective caps.

5.1.2 Level C. Indicators shall be provided sufficient protection against physical and mechanical damage to assure safe delivery without degradation of the item reliability from the supply source to the first receiving activity for immediate use.

5.2 Packing

5.2.1 Level A. Indicators packaged in accordance with 5.1.1 above shall be packed for shipment in exterior type shipping containers conforming to PPP-B-636, water-resistant grade.

5.2.2 Level B. The indicator packaged as specified in 5.1.1 shall be packed for shipment in domestic type exterior containers conforming to PPP-B-636.

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet the consolidated freight classification rules or regulations of other common carriers as applicable to the mode of transportation.

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5.3 Physical protection. Containerization, cushioning, blocking, bracing, bolting, and testing to determine adequacy of the completed package shall be in accordance with the applicable requirements of MIL-P-7936, modified to include the rough handling test for level C.

5.4 Marking. Container marking shall be in accordance with the applicable requirements of MIL-STD-129. The nomenclature shall be as follows:

Indicator, Electrical, Tachometer
ERU-12/A

6. NOTES

6.1 Intended use. The indicator covered by this specification is intended for helicopter use in indicating the rotational speed in rpm of a turbojet shaft engine on the engine scale and the rotational speed of the rotor in rpm on the rotor scale. This tachometer is suitable for use on the UH-1F helicopter but NOT suitable for use on the UH-1B, UH-1D and UH-1E.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Filling medium (see 3.7.3)

c. Dial finish requirements if other than as specified in 3.9

d. Whether sampling plan B or C is to be omitted (see 4.5.2)

e. Levels of packaging and packing required and reinspection date requirements.

6.3 Definitions. For the purpose of this specification, the following definitions will apply.

6.3.1 Tapping. The phrase "gently tapped" is defined as the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality instruments.

6.3.2 Hermetic seal. A hermetic seal is a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets.

6.3.3 Amplitude. Amplitude is defined as the extent of motion as measured from one extreme to the opposite extreme.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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